

Hypothesis Testing

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May 4, 2026

Outline

Recap

Two sample t-test

F-test

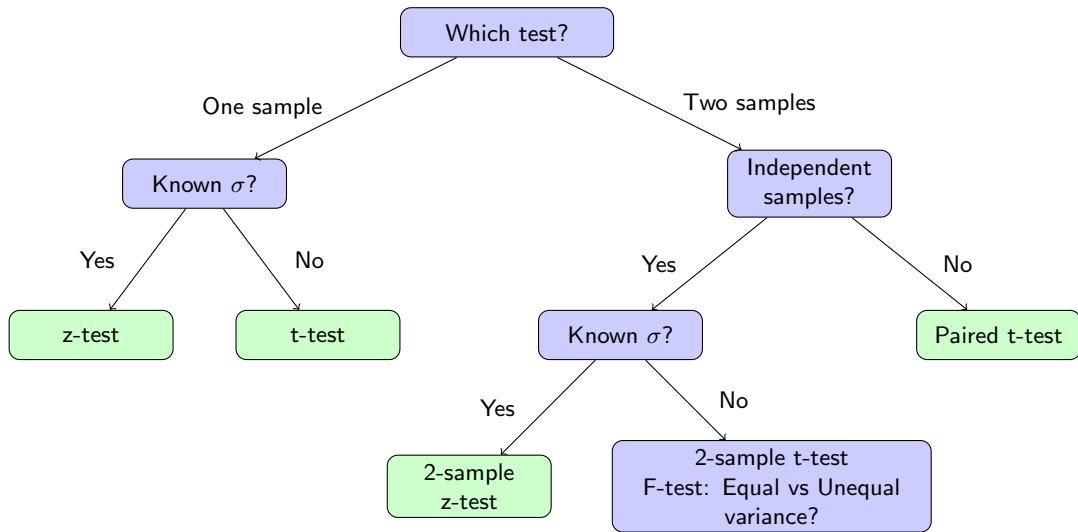
Paired t-test

Lab Report

Procedure for Hypothesis Testing

1. State your null and alternate hypothesis
2. Define significant level α
3. Decide the test statistic (Z, T, F)
4. Find the rejection region (RR)
5. Calculate your test statistic and compare with RR
6. Make a decision whether reject H_0 or not

Types of Hypothesis Testing



One-tailed vs Two-tailed

Two sample t-test

- ▶ Hypothesis: $H_0 : \mu_1 = \mu_2$ vs. $H_A : \mu_1 \neq \mu_2$
- ▶ Test Statistic:
 - ▶ Equal Variance $\sigma_1^2 = \sigma_2^2 = \sigma^2$

$$T = \frac{\bar{x}_1 - \bar{x}_2 - 0}{\sqrt{\frac{s_{pooled}^2}{n_1} + \frac{s_{pooled}^2}{n_2}}} \stackrel{H_0}{\sim} t(n_1 + n_2 - 2), \text{ where } s_{pooled}^2 = \frac{(n_1 - 1)s_1^2 + (n_2 - 1)s_2^2}{(n_1 + n_2 - 2)}$$

- ▶ Unequal Variance $\sigma_1 \neq \sigma_2$

$$T = \frac{\bar{x}_1 - \bar{x}_2 - 0}{\sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}} \stackrel{H_0}{\sim} t(df_{Welch}), \text{ where } df_{Welch} = \frac{(\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2})^2}{\frac{(\frac{s_1^2}{n_1})^2}{n_1 - 1} + \frac{(\frac{s_2^2}{n_2})^2}{n_2 - 1}}$$

- ▶ Rejection Region:

$$RR = \{|T| \geq t_{\frac{\alpha}{2}}(df)\} = \{T \geq t_{\frac{\alpha}{2}}(df) \text{ or } T \leq -t_{\frac{\alpha}{2}}(df)\}$$

Two sample t-test

We get data1 and data2, and we want to test whether $\mu_1 = \mu_2$ under $\alpha = 0.05$

```
data1 <- c(27, 24, 18, 29, 30, 27)
data2 <- c(23, 28, 20, 19, 35, 23)
```

► Equal Variance

```
t.test(data1, data2, mu = 0, var.equal = TRUE, conf.
       level = 0.95)
```

► Unequal Variance

```
t.test(data1, data2, mu = 0, var.equal = FALSE, conf.
       level = 0.95)
```

F-test

- ▶ Hypothesis: $H_0 : \sigma_1^2 = \sigma_2^2$ vs. $H_A : \sigma_1^2 \neq \sigma_2^2$
- ▶ Test Statistic: the ratio of two sample variances

$$F = \frac{s_1^2}{s_2^2} \stackrel{H_0}{\sim} F_{n_1-1, n_2-1, [\alpha/2]}$$

- ▶ Rejection Region:

$$RR = \{F > F_{n_1-1, n_2-1, [\alpha/2]}\}$$

F-test

Assume we get two sample: group1 and group2

We want to test whether the population variance of these two groups are different?

```
# Sample data
group1 <- c(15.2, 16.5, 14.8, 16.9, 15.5, 15.8)
group2 <- c(14.0, 13.8, 15.2, 13.9, 14.1, 14.5)

# Perform the F-Test
var.test(group1, group2)
```

Paired t-test

- ▶ The differences between each pair(i) of sample units

$$d_i = x1_i - x2_i$$

- ▶ Calculate $\bar{d}$ and s_d
- ▶ Hypothesis: The population mean of the differences is zero

$$H_0 : \mu_d = 0 \text{ vs. } H_A : \mu_d \neq 0$$

- ▶ Test Statistic:

$$t = \frac{\bar{d} - 0}{s_d / \sqrt{n}}$$

- ▶ Rejection Region:

$$RR = \{|t| > t_{n-1}\}$$

Paired t-test

We got two groups: group1 and group2. We knew that these two groups are dependent

```
# Sample data
group1 <- c(rnorm(100, mean = 14, sd = 0.3))
group2 <- c(rnorm(100, mean = 13, sd = 0.2))

# Perform the paired t -test
t.test(group1, group2, paired = TRUE)
```

Lab Report

- ▶ The report file will be in the NTU COOL Assignment section
- ▶ Submit your file in pdf format
- ▶ Deadline: 5/10 (Sun) 23:59
- ▶ Late submission is not accepted